

Two-sample tests for relevant differences in persistence diagrams

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Abstract: We study two-sample tests for relevant differences in persistence diagrams obtained from L^p - m -approximable data $(\mathcal{X}_t)_t$ and $(\mathcal{Y}_t)_t$. To this end, we compare variance estimates w.r.t. the Wasserstein metrics on the space of persistence diagrams. In detail, we consider two test procedures. The first compares the Fréchet variances of the two samples based on estimators for the Fréchet mean of the observed persistence diagrams $\text{PD}(\mathcal{X}_i)$ ($1 \leq i \leq m$), resp., $\text{PD}(\mathcal{Y}_j)$ ($1 \leq j \leq n$) of a given feature dimension. We use classical functional central limit theorems to establish consistency of the testing procedure. The second procedure relies on a comparison of the so-called independent copy variances of the respective samples. Technically, this leads to functional central limit theorems for U-statistics built on L^p - m -approximable sample data.

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We study two-sample tests for relevant differences in the (Fréchet) variances of observed persistence diagrams $\text{PD}(\mathcal{X}_i)$ ($1 \leq i \leq m$), resp., $\text{PD}(\mathcal{Y}_j)$ ($1 \leq j \leq n$) which each feature a weak dependence relation. Statistical tests on relevant differences have enjoyed quite some popularity recently. In particular, in the domain of functional time series there are major contributions; we refer to Dette and Wied [15], Dette and Wu [16] and Aue et al. [2]. This testing problem has not received much attention in topological data analysis (TDA) so far.

Classical two-sample tests focus on hypotheses of the form

$$H_0 : T_{\mathcal{X}} = T_{\mathcal{Y}} \text{ vs. } H_1 : T_{\mathcal{X}} \neq T_{\mathcal{Y}},$$

for suitable real-valued statistics $T_{\mathcal{X}} = T((\text{PD}(\mathcal{X}_i))_{i=1}^m)$ and $T_{\mathcal{Y}} = T((\text{PD}(\mathcal{Y}_j))_{j=1}^n)$. In contrast, we study relevant hypotheses of the form

$$H_0 : |T_{\mathcal{X}} - T_{\mathcal{Y}}|^2 \leq \Delta \text{ vs. } H_1 : |T_{\mathcal{X}} - T_{\mathcal{Y}}|^2 > \Delta.$$

The statistics $T_{\mathcal{X}}$ and $T_{\mathcal{Y}}$ which we consider are variances estimates of a sequence of persistence diagrams $\text{PD}(\mathcal{X}_i)$ ($1 \leq i \leq m$), resp., $\text{PD}(\mathcal{Y}_j)$ ($1 \leq j \leq n$). The underlying data generating processes $(\mathcal{X}_t)_t$ and $(\mathcal{Y}_t)_t$ are L^p - m -approximable

processes with values in a suitable subset $\mathfrak{D}$ of $\mathbb{R}^d$. We assume that the diagrams are built with the Čech filtration function which is a classical choice in TDA. We refer to the book of Boissonnat et al. [5] and the introductory article of Chazal and Michel [8] for a background on the subject, in particular, the notion of a filtration function in TDA and the construction of persistence diagrams from simplicial complexes.

Early celebrated contributions which highlighted the potential of TDA are Edelsbrunner et al. [20], Zomorodian and Carlsson [29] and Carlsson et al. [6]. A vast share of recent contributions in TDA focus on computational and practical aspects of TDA, nevertheless the methods have received some attention in the area of (statistical) time series as well, we refer to Seversky et al. [25], Umeda [28].

However, despite the widespread dissemination across various disciplines, for large parts of TDA, the development of mathematically sound statistical tests is still in its infancy. Our contribution to this matter is as follows. We study two two-sample tests. The first is based on Fréchet means and Fréchet variances. Properties of Fréchet variances of a sequence iid data objects have been studied by Dubey and Müller [17, 18], e.g., in the context of structural breaks. The second approach is via so-called independent copy (inco)-variances which are a special case of U-statistics (Hoeffding [21]). Dehling and Wendler [12], Leucht [23] study U-statistics the context of dependent data. Our contribution is as follows: We develop in both settings self-normalized tests for relevant differences in the variances of persistence diagrams which are generated by weakly dependent data. Moreover, we develop a function central limit theorem for general U-statistics of weakly dependent data because these underlying structure for the inco-variance approach.

We conclude with some notation. We will assume that $\mathbb{R}^d$ is equipped with some l^r -norm, i.e., $\|x\| = \|x\|_r = (\sum_{i=1}^d |x_i|^r)^{1/r}$ for some $r \geq 1$. Also for $x, y \in \mathbb{R}^d$ we may write (x, y) to denote the $\mathbb{R}^{2d}$ -valued vector $(x_1, \dots, x_d, y_1, \dots, y_d)$ and – unless otherwise stated – all vector quantities are always viewed as column-vectors. The d -dimensional Lebesgue measure of a Borel set $A \subseteq \mathbb{R}^d$ is denoted by $|A|$. The closed a -ball at a point $x \in \mathbb{R}^d$ is the set $B(x, a) = \{y : \|x - y\|_2 \leq a\}$.

The rest of the manuscript is organized as follows. Section 1 contains a short introduction to the main objects from TDA as well a detailed description of the framework for the two sample tests. The Subsection 1.1 contains the results for two sample tests based on Fréchet means, while the Subsection 1.2 treats the case for the inco-variances.

The mathematical background on U-statistics is treated in Section 2. All mathematical details are deferred to Section 3.

1. Tests for persistence diagrams

Throughout the article, we assume that the underlying simplicial complexes are all built with the Čech filtration function. That is, given a finite $P \subseteq \mathbb{R}^d$, a simplex $\{x_0, \dots, x_k\} \subseteq P$ is an element of the Čech complex at parameter $a > 0$

if and only if $\bigcap_{i=0}^k B(x_i, a) \neq \emptyset$. We write $\mathcal{C}(P, a)$ for this simplicial complex.

Let $\mathcal{D}$ denote the set of all persistence diagrams and consider the L^p -Wasserstein metrics

$$W_r(U, V) = \inf_{\gamma} \left(\sum_{u \in U} \|u - \gamma(u)\|_{\infty}^r \right)^{1/r}, \quad r \in [1, \infty),$$

$$W_{\infty}(U, V) = \inf_{\gamma} \sup_{u \in U} \|u - \gamma(u)\|_{\infty}.$$

W_{∞} is also known as the bottleneck distance W_B . Then, it is true that $W_{\infty}(U, V) \leq W_q(U, V) \leq W_r(U, V) \leq \max\{\#U, \#V\}^{1/r-1/q} W_q(U, V)$ for $1 \leq r \leq q \leq \infty$ by the usual norm inequalities in the Euclidean space.

Define for $r \in [1, \infty]$ the sets

$$\mathcal{D}_r := \{U \in \mathcal{D} \mid W_r(U, U_{\emptyset}) < \infty\},$$

where $U_{\emptyset} = \{(x, x) : x \geq 0\}$ is a diagram with just the diagonal. Then $(\mathcal{D}_r, W_r)$ is a complete metric space and a set $S \subset \mathcal{D}_r$ is totally bounded if and only if it is bounded, off-diagonally birth-death bounded and uniform, see Mileyko et al. [24][Theorem 6 and 21] for more details, in particular the notion of ‘uniform’. Moreover, it is known that $(\mathcal{D}_2, W_2)$ is a non-negatively curved Alexandrov space for $p = 2$, see Turner et al. [27][Theorem 2.5].

We equip the metric space $(\mathcal{D}_r, W_r)$ with its Borel- σ -field $\mathcal{B}(\mathcal{D}_r)$. To any a probability measure ρ on $(\mathcal{D}_r, \mathcal{B}(\mathcal{D}_r))$, there corresponds a so-called Fréchet function

$$F_{\rho} : \mathcal{D}_r \rightarrow \mathbb{R}, \quad V \mapsto \int_{\mathcal{D}_r} W_r^2(U, V) \rho(dU). \quad (1.1)$$

The *Fréchet variance* $\text{Var}(\rho) \in \mathbb{R}$ and the *Fréchet mean(s)* $\mu(\rho) \subseteq \mathcal{D}_r$ of ρ are then defined as the solutions of the optimization problem

$$\text{Var}(\rho) = \inf_{V \in \mathcal{D}_r} F_{\rho}(V), \quad \mu(\rho) = \arg \min_{V \in \mathcal{D}_r} F_{\rho}(V). \quad (1.2)$$

If the probability measure ρ is induced by a $\mathcal{D}_r$ -valued random element X , the corresponding Fréchet function can simply be expressed via an expectation

$$F_X : \mathcal{D}_r \rightarrow \mathbb{R}, \quad V \mapsto \mathbb{E}[W_r^2(X, V)]. \quad (1.3)$$

We may write $\text{Var}(X)$ respectively $\mu(X)$ for the solutions of (1.2) and refer to these quantities as Fréchet variance respectively Fréchet mean(s) of X . The notational similarity to the Euclidean mean and variance is of course intentional, since the Fréchet mean and variance poses a coherent generalization to metric spaces. Especially, the Fréchet variance provides a natural measure for dispersion.

Under mild assumptions on ρ (e.g., bounded support), respectively X , existence of a Fréchet mean is guaranteed, we refer to Mileyko et al. [24] [Theorem 21 and Lemma 27] for a detailed study on Fréchet means of persistence diagrams.